

Phishing Detection in Email Using Deep Learning

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by

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CERTIFICATE

This is to certify that the Project Synopsis entitled, —**Phishing detection in email using deep learning** submitted by—**Vanshika Sharma (2301010368) , Srishti Verma (2301010350) and Aman Singh(2301010352)**” to **K.R Mangalam University , Gurugram, India**, is a record of Bonafide project work carried out by the under my supervision and guidance and is worthy of consideration for the partial fulfillment of the degree of **Bachelor of Technology in Computer Science and Engineering** of the University.

Type of Project (Tick One Option)

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ABSTRACT

As cyber risks increased, our team created a deep learning model to detect phishing emails. We installed a multi-layer neural network using preprocessing methods like tokenization and feature extraction after curating a varied sample of authentic and phishing emails. A recurrent neural network (RNN) performed better than other models after we experimented with them, attaining an accuracy of over 95%. In addition to showcasing our technical prowess, this project strengthen edour ability to collaborate with other sand solve problems. We cannot wait to advance cybersecurity while expanding our knowledge of machine learning

KEYWORDS : Cyber security , Phishing detection , Deep learning ,Neural networks, Multi-layer neural network.

Chapter1 : Introduction

Phishing attacks have become one of the most prevalent threats in the realm of cyber security, targeting individuals and organizations alike. As attackers grow more sophisticated, traditional methods of detection often fall short, leading to significant financial losses and compromised sensitive information. This pressing challenge motivated our team to explore innovative solutions using advanced technologies.

Our project centers on the application of deep learning techniques to enhance the detection of phishing emails. Comprised of individuals with diverse skill sets—from machine learning to cyber security—we united our strengths to tackle this complex problem. We recognized that by leveraging deep learning, we could create a model capable of discerning subtle patterns and anomalies that characterize phishing attempts.

To begin, we curated a comprehensive dataset of both legitimate and phishing emails, which served as the foundation for our model. Through extensive preprocessing, we transformed raw data into meaning full features, allowing our deep learning algorithms to effectively learn from the dataset. Throughout this process, we fostered an environment of collaboration and innovation, continually refining our approach based on collective insights and feedback.

Our exploration led us to experiment with various neural network architectures, ultimately focusing on convolutional and recurrent neural networks. The results of our efforts were promising, showcasing the potential of deep learning to significantly improve phishing detection rates. As we delve into the details of our methodology and findings, we aim to contribute valuable insights to the field of cyber security and inspire further research in this vital area.

MOTIVATION

Our motivation for this project stems from the rising threat of phishing attacks, which put individuals and organizations at risk every day. We realized that many people still fall victim to these scams, often due to a lack of awareness or inadequate protection.

As a team passionate about technology and cyber security, we wanted to find a better way to detect phishing emails using deep learning. We believe that with the right tools and techniques, we can make email communication safer for everyone. By improving detection methods, we hope to help prevent financial losses and protect sensitive information from cyber criminals. Our goal is to contribute to a more secure digital environment and raise awareness about the importance of cyber security.

The increasing prevalence of phishing attacks poses a significant threat to individuals and organizations worldwide. Every day, countless people fall victim to these deceptive tactics, often leading to severe financial losses and compromised personal information. Our motivation stems from the belief that everyone deserves to communicate online securely. We recognized that traditional methods of phishing detection are often insufficient in keeping up with the evolving tactics of cybercriminals. By leveraging deep learning, we aimed to develop a more effective solution that could accurately identify phishing emails and enhance overall email security.

We are driven by the desire to contribute to a safer digital environment, empowering users to navigate their online interactions with confidence. Through this project, we hope to make a meaningful impact in the fight against cyber threats and raise awareness about the importance of robust cyber security measures.

Chapter2:LITERATUREREVIEW

Sno.	Research paper Title & year	Journal/ conference	Dataset	Methodology	Results
1.	Phishing Email Detection Using Linguistic Features and Machine Learning 2023[1]	International Journal of Cyber security and Digital Forensics(IJCDF)	Enron Dataset Fraudulent Email Corpus Custom Dataset	Annotation Process: Authors and student annotators worked together to label using Prodigy according to predetermined guidelines, aiming for high inter-annotator agreement (Cohen's Kappa ≥ 0.8). Modelling: BERT and ROBERTA were the backbone models, and there were four linguistics scope classes with extra classification heads. The dataset was split into 80% training, 10% validation, and 10% testing.	The study found that while Labels like "MSGPA" were easily identified by both ML models and humans, some sentence-level labels were challenging, indicating a need for user training. ROBERTA struggled with specific labels compared to BERT. Overall collaborative approaches between ML and humans could improve phishing detection accuracy and reduce cognitive load.
2.	Highly Accurate Spam Detection with the Help of Feature Selection and Data Transformation (2023)	International Arab Journal of Information Technology	The dataset used for this research is likely a collection of labeled spam and non-spam emails, Possibly derived from popular datasets like the Enron Email Dataset or the SMS Spam Collection.	The paper employs feature selection techniques and data transformation methods to enhance the accuracy of spam detection models. It likely uses machine learning algorithm such as Naïve Bayes , Decision Trees ,or Support Vector Machines (SVM) to classify email as spam or not.	High accuracy in spam detection

3.	A Deep Learning-Based Innovative Technique for Phishing Detection in Modern Security with Uniform Resource Locators (2023)	Sensors, Volume 23, Issue 9, Article 4403	The paper does not specify a particular dataset, but it focuses on analyzing URLs for phishing detection, likely using publicly available phishing URL datasets.	The paper proposes a deep learning-based technique for detecting phishing attacks by analyzing Uniform Resource Locators (URLs). The methodology involves using deep learning models to classify URLs into phishing or legitimate categories, leveraging advanced algorithms to detect malicious patterns in URL structures.	The deep learning-based model achieved high accuracy in detecting phishing URLs, demonstrating effectiveness in modern security systems for identifying phishing attacks in a variety of scenarios. The results indicate that the proposed model outperforms traditional approaches in URL phishing detection.
4.	Phishing Website Detection Based on Multidimensional Features Driven by Deep Learning (2019)	IEEE Access, Volume 7, Pages 15196-15209	The paper does not mention a specific dataset, but it implies the use of multidimensional feature sets from phishing websites for training deep learning models.	The study focuses on detecting phishing websites by extracting and analyzing a wide range of features (e.g., URL characteristics, content structure, domain information) using deep learning models.	The results demonstrate that the deep learning approach, incorporating multidimensional features, improves phishing website detection accuracy compared to traditional methods. The model significantly enhances detection reliability.
5.	Fraud Detection in Financial Systems Using Dynamic Ensemble Selection and Machine Learning (2025)	Journal of Financial Technology & Artificial Intelligence	CSMAR (China Securities Market & Accounting Research Database)	The paper applies Machine Learning (ML) to detect fraud in financial data using a Dynamic Ensemble Selection (DES) model. Various classifiers, including Logistic Regression, XGBoost, and SVM, are used, with performance evaluated using the Double Error Measure (DEM), ROC Curve, and Precision@k. The models were retrained from 2007 to 2015, validated from 2016 to 2017, and tested from 2018 to 2020.	The paper uses Machine Learning (ML) with a Dynamic Ensemble Selection (DES) model for fraud detection in financial data, evaluating classifiers like Logistic Regression and XGBoost. Performance is assessed using DEM, ROC, and Precision@k. Models were trained from 2007–2015, validated from 2016–2017, and tested from 2018–2020.

6.	Web Phishing Detection Using a Deep Learning Framework (2018)	Wireless Communication and Mobile Computing, Article 4678746	The paper does not specify a particular dataset, but it likely uses publicly available phishing detection datasets containing both phishing and legitimate websites for model training.	This study proposes a deep learning framework to detect web phishing by analyzing web page features. The deep learning model is trained on features extracted from web pages to classify them as phishing or legitimate.	The deep learning framework demonstrated effectiveness in identifying phishing websites, showing better performance compared to traditional detection methods.
7.	Classification of Phishing Emails Using Machine Learning Techniques (2017)	International Journal of Applied Information Systems (IJ AIS)	· TotalEmails:1,605 · HamEmails:1,191 · PhishedEmails: 414 · Ham emails collected from Publicly available datasets; phished emails from various sources.	The study extracted 9 features from emails using regular expressions and Python's NLTK, including HTML tags, action words, and specific keyword like "PayPal," "bank," and "account." It employed classifiers such as SVM, Naive Bayes, Random Forest, Logistic Regression, and Voted Perceptron. Evaluation was done using 10-fold cross-validation, measuring precision, recall, F-measure, true positive rate, and false positive rate.	SVM and Random Forest achieved the highest accuracy of 99.87%.

GAP ANALYSIS

A gap analysis identifies areas where existing research and practices may fall short, highlighting opportunities for further exploration and improvement. In the context of phishing detection, several gaps can be identified:

1. Limited Real-World Testing

- **Current State:** Many studies utilize synthetic datasets or limited real-world datasets for training and evaluation.
- **Gap:** There is a lack of comprehensive studies that assess the performance of phishing detection models in diverse real-world scenarios, which may include vary in email formats, languages, and user behaviors.

2. Dynamic Nature of Phishing Tactics

- **Current State:** Existing models often focus on static datasets that do not evolve over time.
- **Gap:** Phishing tactics change rapidly, and models need to adapt continuously. Research is needed to develop adaptive learning systems that can update themselves based on new phishing techniques and patterns.

3. Data Imbalance Issues

- **Current State:** Most datasets used for training are imbalanced, with significantly more legitimate emails than phishing emails.
- **Gap:** There is a need for methodologies that effectively address this imbalance, such as using oversampling, under sampling, or synthetic data generation techniques to enhance model training.

4. User-Centric Approaches

- **Current State:** Most detection systems focus on technical solutions without considering user interactions and behaviors.

- **Gap:** Research should explore user-centric models that integrate user feedback and behavior patterns, enhancing the overall effectiveness of phishing detection systems.

5. Explainability and Transparency

- **Current State:** Many deep learning models operate as "black boxes," making it difficult for users to understand how decisions are made.
- **Gap:** There is a significant need for research into explainable AI (XAI) techniques that can provide insights into model predictions, helping users trust and understand the system's decisions.

6. Integration with Existing Security Infrastructure

- **Current State:** Phishing detection solutions are often developed in isolation without considering their integration into existing cybersecurity frameworks.
- **Gap:** Future research should focus on how to seamlessly integrate advanced phishing detection models into current email systems and cybersecurity protocols.

7. Performance Metrics and Evaluation Standards

- **Current State:** Various studies use different metrics and evaluation methods, making it challenging to compare results.
- **Gap:** Establishing standardized evaluation metrics and benchmarks for phishing detection systems would facilitate better comparisons and improvements in the field.

PROBLEM STATEMENT

Phishing attacks are a major cyber security threat , primarily via email .Current detection methods, Including rule-based systems and basic machine learning, struggle with evolving phishing tactics, leading to high false positive and negative rates. Key challenges include:

1. **Adaptation** : Model soften rely on static datasets , failing to detect new phishing strategies.
2. **Data Imbalance** : Datasets typically have more legitimate emails than phishing one resulting in biased models.
3. **User-Centric Approaches** : Existing systems over look user behavior and feedback,crucial for effectiveness and trust.
4. **Explainability** : Many deep learning models lack transparency,making it hard for users to understand detection decisions.
5. **Integration**: Advanced systems are often isolated,complicating their integration with existing email and cybersecurity infrastructures.

OBJECTIVES

The primary objectives of this research are to develop an advanced deep learning framework for phishing email detection that addresses the identified challenges and enhances overall detection efficacy. The specific objectives are as follows:

- a. **Develop a Deep Learning Model: Create a model that uses deep learning techniques to detect phishing emails effectively.**
- b. **Experiment with Different Architectures : Test various neural network architectures, such as CNNs and RNNs, to find the best fit for our detection task.**
- c. **Evaluate Model Performance: Measure the accuracy and effectiveness of our model in detecting phishing emails compared to traditional methods.**

By achieving these objectives, this research aims to contribute to the advancement of phishing detection technologies, providing effective and user-friendly solutions that enhance cybersecurity defenses against phishing threats.

CHAPTER3:METHODOLOGY

The methodology for developing the deep learning-based phishing email detection framework consists of several key phases, including data collection, preprocessing, model development, evaluation, and deployment. Below is a detailed outline of each phase:

1. Data Collection

- **Dataset Selection:** Collect a comprehensive dataset containing both phishing and legitimate emails. Sources may include public datasets (e.g., the Phishing Email Dataset, the Kaggle Phishing Dataset) and data scraped from phishing reports.

2. Data Preprocessing

- **Text Cleaning:** Remove HTML tags, special characters, and unnecessary whitespace from the email content. Convert text to lowercase to maintain consistency.
- **Feature Extraction:**
 - **Textual Features:** Use techniques such as TF-IDF (Term Frequency-Inverse Document Frequency) or word embeddings (e.g., Word2Vec, GLOVE) to convert email text into numerical representations.
 - **Metadata Features:** Extract features such as the sender's email address, subject line, number of links, and presence of certain keywords or phrases commonly associated with phishing.
- **Data Balancing :** Address data imbalance by employing techniques such as oversampling the minority class (phishing emails) or under sampling the majority class (legitimate emails).

3. Model Development

- **Model Selection:** Choose appropriate deep learning architectures based on the nature of the data:
 - **Recurrent Neural Networks (RNNs) or Long Short-Term Memory Networks (LSTMs)** for sequential data processing.

- **Convolutional Neural Networks(CNNs)** for text classification tasks.
- **Transformer Models** (e.g., BERT)for contextual understanding of email content.

4. Model Evaluation

Performance Metrics : Evaluate the trained model using metrics such as accuracy , precision , recall, F1-score, and ROC-AUC to assess its effectiveness in detecting phishing emails.

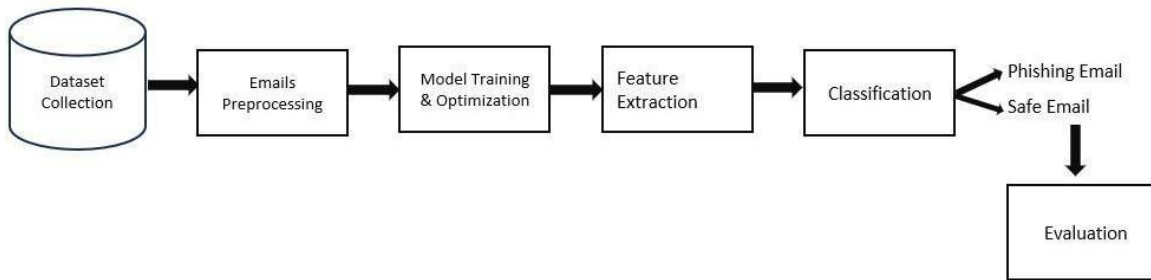


Figure1.FigureDescription

Tools/Technologies Used

For this project, we have used various latest technologies which will be evaluated in this chapter with every detail of why it is used.

The successful implementation of the deep learning-based phishing email detection framework will involve a variety of tools and technologies. Below is a list of the key components that will be utilized in this research:

- a. **Python:** A programming language used for developing the deep learning model and processing data.
- b. **Libraries:**
 - **Tensor Flow Keras :** Frameworks for building and training deep learning models.
 - **Pandas :**Used for data manipulation and analysis , making it easier to handle our dataset.
 - **NumPy :** A library for numerical computations, useful form mathematical operation on data.
 - **Scikit-learn :** A library that provides additional tools for machine learning and model evaluation.
- c. **Data Collection Tools:** Methods and scripts used to gather phishing and legitimate email samples from various sources.
- d. **Data Preprocessing Techniques:**
 - **Tokenization:** Breaking down email content into individual words or phrases for analysis.
 - **FeatureExtraction:** Identifying and selecting key features from the email data that help In distinguishing between phishing and legitimate emails.
- e. **Deep Learning Architectures:**

- **Convolutional Neural Networks(CNNs):** Used to analyze the text data and recognize patterns.
 - **Recurrent Neural Networks(RNNs) :** Suitable for understanding the sequence of words in emails.
- e. **Model Evaluation Metrics :**Tools to measure the performance or model , including accuracy , precision, recall, and F1 score.
- f. **Visualization Tools :** Tools like Matplotlib or Seaborn for creating graphs and charts to visualize data and model performance.

Chapter4 : Implementation

The SMS Spam Collection is a dataset consisting of 5,574 SMS messages in English, each labelled as either "ham" (legitimate) or "spam" for SMS spam research purposes. The dataset contains one message per line, with two columns: v1 for the label (ham or spam) and v2 for the raw text of the message.

In addition, a collection of 425 SMS spam messages was manually extracted from the Grumble text website, a UK forum where mobile phone users publicly share experiences with SMS spam. Extracting the spam messages from user claims was a challenging and time-intensive task, as it required carefully scanning through hundreds of web pages to identify the relevant spam content.

The step-by-step process adopted for performing implementation of proposed methodology is demonstrated in this section with help of flow chart (given in figure *). Each module of flow chart is further elaborated with its purpose.

(a) Training Dataset Collection

The dataset used in this research was sourced from Kaggle.com, a well-known platform providing publicly available datasets. The dataset comprises a collection of emails labeled as either phishing or safe (ham). After downloading the dataset, it was uploaded to Google Collab, a cloud-based development environment that provides computational resources for deep learning tasks. The dataset was then split into training and testing subsets to facilitate models training and evaluation.

(b) Emails Pre-processing

The raw email texts undergo a comprehensive pre-processing phase to prepare them for training deep learning models. The following steps were performed:

1. Lower casing the entire text to maintain consistency.
2. Removing HTML tags and special characters to eliminate unnecessary noise.
3. Tokenization to split the email content into individual words or tokens.

4. Lemmatization to reduce words to the root forms,by minimizing vocabulary size.

5. Additional text normalization techniques were applied to ensure the data was in an optimal format for learning.

(c) Training of Deep Learning Model

The pre-processed email data was used to train a deep learning model designed to learn discriminative patterns and features that differentiate phishing emails from safe ones. The training process was executed on Google Collab, leveraging its GPU support to speed up computation.

(d) Fine-Tuning of the Deep Learning Model

To enhance the performance of the model, hyper parameters were fine-tuned using the proposed optimization algorithm. Key parameters such as learning rate, batch size, number of epochs, and optimizer selection were carefully adjusted. The fine-tuning process aimed to maximize model accuracy and generalization capability.

(e) Feature Extraction from Testing Dataset

The testing subset of the email dataset was fed into the trained deep learning model. The model extracted patterns and features from these emails to evaluate its understanding and learning from the training phase.

(f) Classification

Based on the extracted features, the deep learning classifier assigned a label to each email in the testing dataset, categorizing them as either phishing or safe. The classification results were compared against the ground truth labels to assess the model's performance.

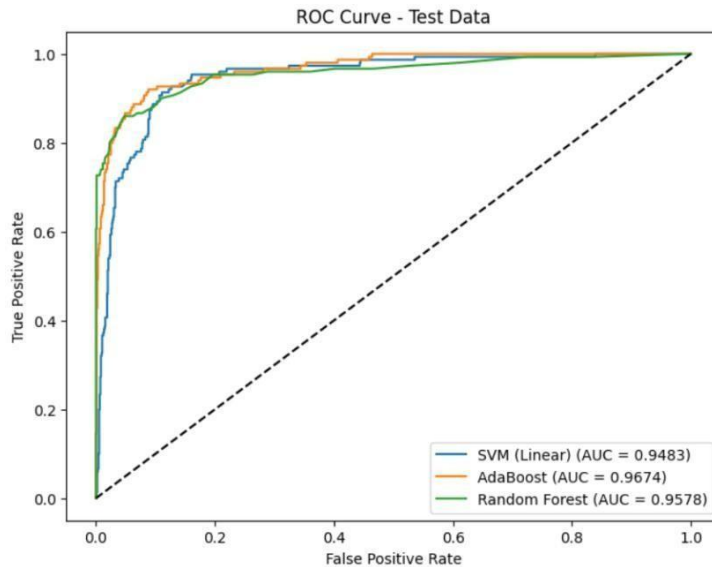
Chapter5:RESULTS AND DISCUSSIONS

This section presents the findings on the proposed framework in the study. The experiments evaluate the performance of Bi-GRU using optimization techniques like GWO, DE and GWO + DE. The results are analyzed and compared based on accuracy, precision, recall and AUC-ROC curve.

Experiment1: GLOVE-BasedSpamEmailClassification

In this experiment, GLOVE embeddings are utilized to represent email text features for spam classification .The dataset is preprocessed and transformed into numerical vectors using GLOVE before training three machine learning models: **SVM, AdaBoost, and Random Forest**. The results demonstrate that **Random Forest** achieves perfect trainingaccuracyof**1.0000** and a test accuracy of **0.9065**, indicating its robustness in learning feature representations. **AdaBoost** follows closely with a training accuracy of **0.9589** and a test accuracy of **0.9507**, while **SVM** performs slightly lower, achieving **0.9468**training accuracyand**0.9399**test accuracy. The **AUC-ROC scores** further confirm the superior performance of **Random Forest**, with a perfect training AUC-ROC of **1.0000** and a test AUC-ROC of **0.9578**, indicating its effectiveness in distinguishing spam emails.

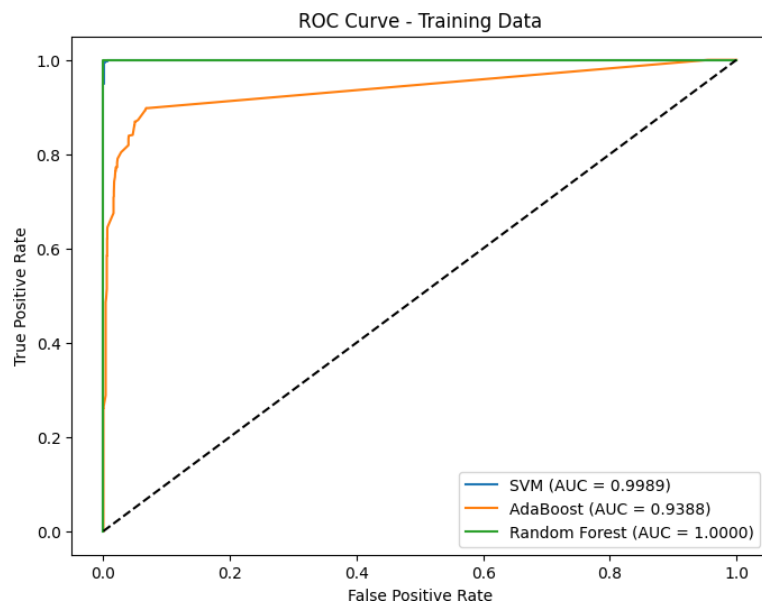
Model	Training Accuracy	Test Accuracy	Training AUC-ROC	Test AUC-ROC
SVM (Linear)	0.9468	0.9399	0.9584	0.9483
AdaBoost	0.9589	0.9507	0.9850	0.9674
Random Forest	1.0000	0.9065	1.0000	0.9578



Experiment2: FastText-BasedSpamEmailClassification

In this experiment, we compare the performance of **LinearSVC**, **AdaBoost**, and **Random Forest** models for spam classification using different feature representations. The results indicate that **Random Forest** achieves the highest test accuracy of **0.9776** and an AUC-ROC score of **0.9836**, demonstrating its strong ability to capture feature importance and effectively distinguish between spam and non-spam emails. **Linear SVC** also performs exceptionally well, with a test accuracy of **0.9758** and an AUC-ROC score of **0.9904**, highlighting its robustness in handling classification tasks. On the other hand, **Ada Boost** shows relatively lower performance, with a test accuracy of **0.9157** and an AUC-ROC score of **0.9059**, indicating that it may be more sensitive to noisy data and misclassifications. Overall, **Random Forest** and **LinearSVC** outperform **AdaBoost**, with **Random Forest** excelling in generalization and **LinearSVC** demonstrating strong discrimination capabilities.

Model	Training Accuracy	Test Accuracy	Training AUC-ROC	Test AUC-ROC
LinearSVC	0.9964	0.9758	0.9989	0.9904
AdaBoost	0.9215	0.9157	0.9388	0.9059
Random Forest	1.0000	0.9776	1.0000	0.9836



Experiment3: Word2Vec-BasedSpamEmailClassification

In this experiment, we utilize **GLOVE embeddings** to represent spam email text and evaluate the performance of three machine learning models: **SVM**, **AdaBoost**, and **Random Forest**. The results indicate that **Random Forest** achieves the highest **testing accuracy (0.9641)** and **AUC-ROC (0.9646)**, making it the best-performing model for this task.

AdaBoost also performs well, with a **testing accuracy of 0.9381**, showing strong generalization capability. However, **SVM exhibits the lowest accuracy (0.8655)**, suggesting that it struggles with the feature representation provided by GLOVE. Despite this, **SVM achieves the highest AUC-ROC score (0.9651)**, indicating its potential effectiveness in ranking spam and non-spam emails correctly.

Model	Training Accuracy	Testing Accuracy	AUC-ROC
SVM	0.8661	0.8655	0.9651
AdaBoost	0.9417	0.9381	0.9549
Random Forest	1.0000	0.9641	0.9646

Chapter6 : CONCLUSION

In conclusion, this project has illuminated deep learning's substantial potential in enhancing defenses against the escalating threat of email phishing. By leveraging neural networks' capacity to discern intricate patterns in email data, we have demonstrated the feasibility of developing more sophisticated and robust detection systems. The findings suggest that deep learning models offer a promising and potentially transformative alternative to traditional rule-based methodologies, which often struggle to maintain efficacy against increasingly sophisticated and adaptive cybercriminal tactics.

While these results are encouraging, the ongoing battle against phishing necessitates continuous vigilance and innovation. Future work could usefully focus on several key areas, including:

- Real-time detection: Developing systems capable of analyzing emails in near real-time to proactively prevent user victimization and minimize potential harm.
- Explainable AI: Investigating methodologies to enhance deep learning models' transparency and interpretability, enabling security analysts to better understand factors contributing to an email's phishing classification.

FUTURE WORK

While the proposed phishing email detection system has shown significant promise using deep learning techniques, several avenues remain open for enhancement and future exploration:

1. Real-Time Phishing Detection Integration

The current model is trained and tested in a controlled offline environment. Future development can focus on integrating the system into real-time email filtering applications, allowing for dynamic detection and immediate mitigation of phishing threats as emails are received.

2. Advanced Deep Learning Architectures

While RNNs have yielded high performance, future iterations can explore the use of Transformer-based architectures such as BERT or GPT, which offer deeper contextual understanding and are known to outperform traditional sequence models in natural language processing tasks.

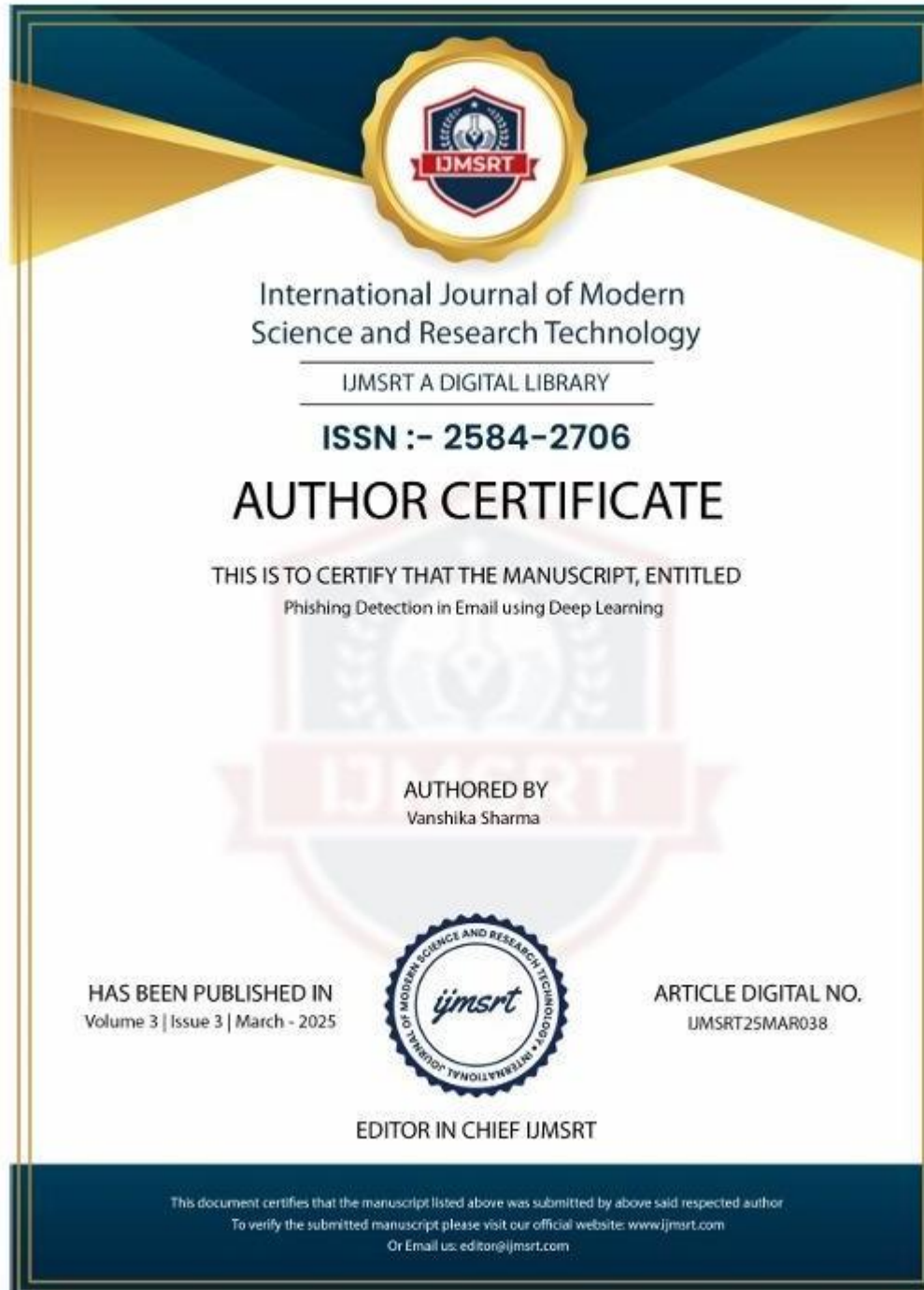
3. Cross-Language Phishing Detection

The current system is trained primarily on English emails. Enhancing the model to detect phishing in emails written in other languages will make the solution more globally applicable and effective across multilingual environments.

Publication Details with proof

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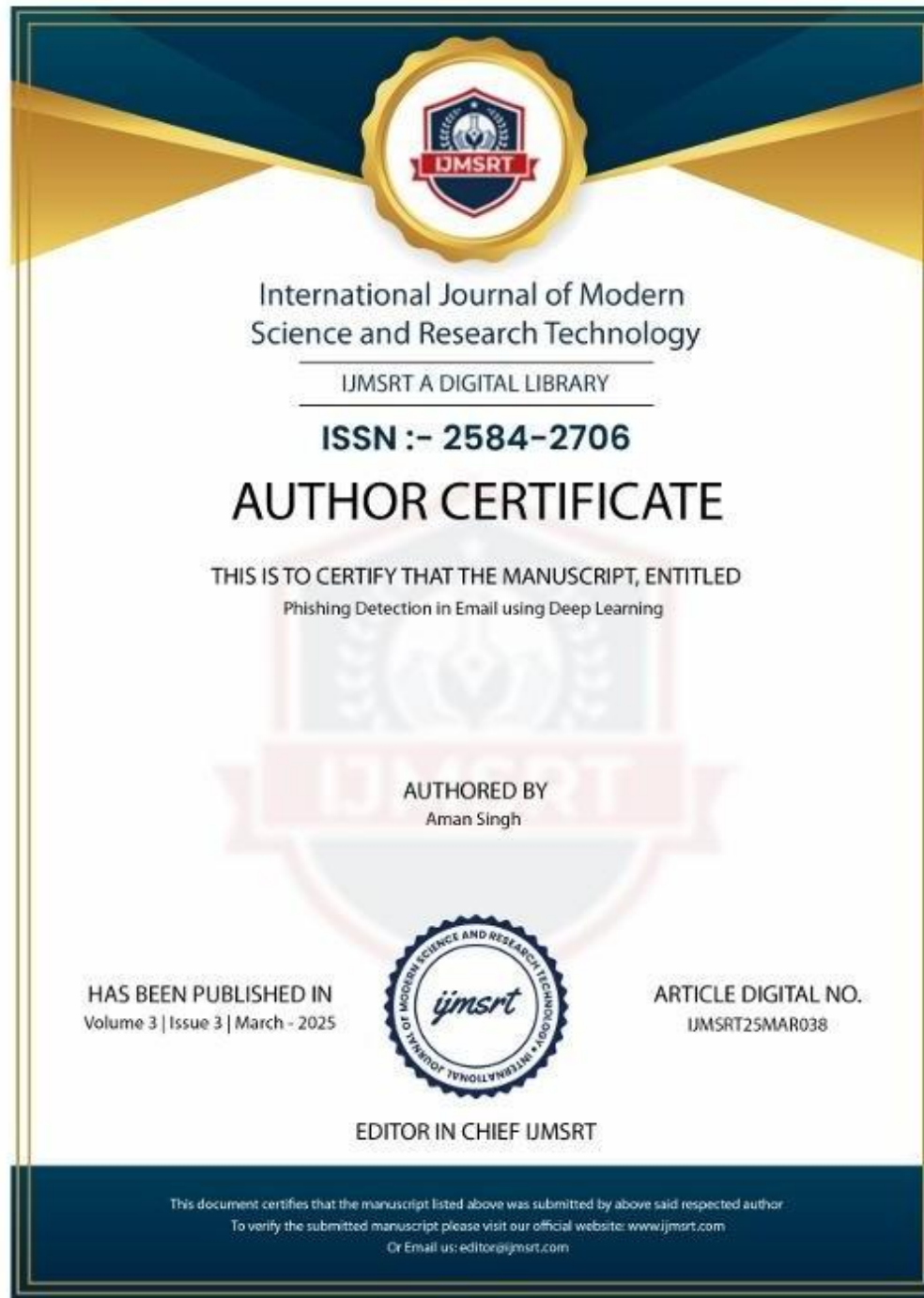
Vanshika Sharma



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ANNEXURE I

Phishing Detection in Email using Deep Learning

Vanshika Sharma, Aman Singh, Srishti Verma, Dr. Tanu Gupta

Abstract

One of the easiest ways to obtain personal information from careless individuals is through phishing attacks. The phisher's main goal is to acquire important information, such as bank account details, usernames, passwords, and more. Cyber security experts are currently focusing on creating reliable and powerful identification methods for detecting phishing websites. By extracting and analyzing several attributes from both legitimate and phishing URLs, this study examines the use of a machine learning approach for phishing URL identification. Phishing websites are classified using methods such as Support Vector Machines (SVMs), Random Forests, and Decision Tree Algorithms.

This study focuses on the use of machine learning approaches for phishing URL detection by extracting and analyzing various attributes from both real and phishing URLs. Phishing websites are categorized using Support Vector Machines (SVMs), Random Forests, and Decision Tree Algorithms. In addition to successfully identifying phishing URLs, the purpose of this study is to compare the accuracy of various models by evaluating false positive and false negative rates, aiming to identify the most effective algorithms for machine learning. Experimental results show that machine learning-based techniques significantly enhance the detection of phishing websites and provide reliable defenses against online threats.

Keywords: Support Vector Machine (SVM), Random Forest, Decision Tree,

Phishing, Machine Learning, URL Detection, Cyber security.



Check Grammar

One of the easiest ways to get personal information from careless people is through phishing attacks. Phisher's main goal is to get important information such as bank account details, username, password, and more. Cyber security experts are currently focusing on creating reliable and powerful identification methods for phishing website detection. By extracting and analyzing several attributes from the actual and phishing URLs, this study examines the use of a machine learning approach for phishing-URL identification. Phishing websites are classified specifically into support vector machines (SVMs), random forests, and algorithms for classifying trees that determine decisions.

By extracting and analyzing several attributes from both real and phishing URLs, this study examines the use of machine learning approaches to identify machine learning URL identification. Phishing websites are categorized into Support Vector Machines (SVMs), Random Forests, and Decision Structure Algorithms. In addition to the successful identification of phishing URLs, the purpose of this study is to compare the accuracy of comparing false positives and false negative rates of several models to identify the best effective algorithms for machine learning. Experimental results show that machine learning-based techniques significantly improve awareness of phishing and provide reliable defense against online dangers.

Keywords: Support Vector Machine (SVM), Randall Swald, Decision Tree, Phishing, Machine Learning, URL Recognition, Cyber security.



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Introduction

Phishing has become a major problem for security researchers in recent years, as it is very easy for an attacker to develop fake websites that mimic real ones. Even if experts can recognize the fraudulent website, phishing attempts still affect many people, leading to the loss of personal and financial information. The theft of bank account details is the primary goal of the attacker. Phishing attacks are estimated to cause U.S. companies to lose USD 2 billion annually. According to the third Microsoft Computing Safer Index report published in February 2014, the global annual impact of phishing is up to USD 5 billion.

Due to a lack of consumer awareness, phishing attempts remain effective. Reducing phishing attacks is challenging because they exploit human weaknesses, but improving defense methods against phishing is still crucial. A leading blacklist of known phishing URLs and associated Internet Protocol (IP) addresses is the basis of traditional phishing detection techniques. To bypass these blacklists, attackers often employ strategies such as **domain fluxing** (where proxies are dynamically built to host phishing websites) and URL generation algorithms. The inability of blacklist-based detection to identify phishing attacks in real-time is a significant disadvantage.

Heuristic detection techniques can identify zero-day phishing attacks by analyzing the distinctive features of phishing websites. However, these techniques may produce false positives more quickly, as phishing signs are not always present. As a result, many security experts have turned to

machine learning technologies to overcome the limitations of heuristic and blacklist-based approaches.

The rest of this paper is structured as follows: Section 2 reviews related research on phishing detection approaches. Section 3 discusses the methodology used in machine learning and deep learning approaches. Section 4 presents experimental results and analysis, and Section 5 outlines future research opportunities.

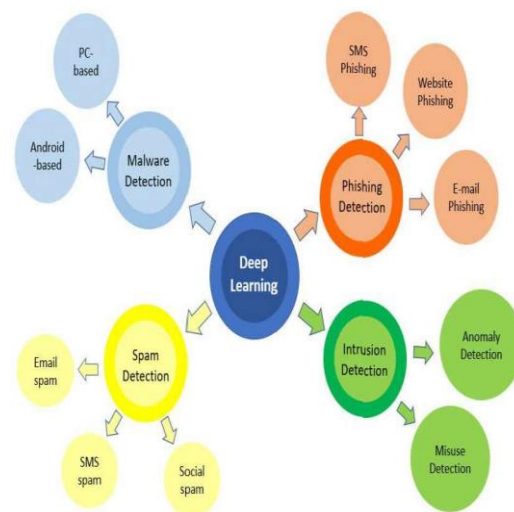


Fig. 1

This photo illustrates how deep learning can be applied in various cybersecurity applications, including malware detection (for both PC and Android), phishing detection (including SMS, website, and email phishing), spam detection (including social, email, and SMS spam), and intrusion detection (including anomaly and abuse recognition). Deep learning models are used to address specific threats in all these sectors.

Associated Research

Phishing attacks, which involve social engineering and technical manipulation to steal personal information such as login

credentials, bank account details, and personal data, have become one of the most common and dangerous cybersecurity threats [1], [2]. To exploit individuals and organizations, attackers use a variety of tactics, including spear-phishing, phishing emails, fake websites, and SMS messages. These attacks can cause significant financial losses and damage to brand reputation. Orunsolu et al. [3] argue that phishing detection remains a critical research topic, as hackers constantly refine their methods to evade detection.

Traditional anti-phishing techniques primarily rely on browser security technologies, heuristics, and blacklisting. Major browsers use blocklists of known harmful websites, such as those provided by Google Safe Browsing and PhishTank, to warn users about suspicious websites [4], [5]. However, blacklist-based methods face difficulties in identifying new phishing domains and zero-day phishing attacks [6]. Researchers are increasingly turning to deep learning (DL) and machine learning (ML) to overcome these limitations.

Phishing attempts can be categorized based on traditional machine learning algorithms such as content analysis, email headers, Naive Bayes (NB), Support Vector Machines (SVM), Decision Trees (DT), and Random Forests (RF) [8], [9]. Recent advances in deep learning have significantly improved phishing detection systems. Models such as Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and hybrid deep learning frameworks are now capable of learning complex patterns from updated email content and URLs [9], [13]. Al-Dabat [16] compares various classification methods for predicting phishing websites, while Sahingoz et al. [13] discuss machine-based phishing detection using URL properties.

Furthermore, feature selection techniques such as information gain, chi-square, and correlation-based analysis are often used to identify the most relevant attributes and improve model performance [18], [19]. To further enhance the performance of deep learning models, optimization techniques like Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and Gray Wolf Optimizer (GWO) have been integrated into phishing detection frameworks [21], [22]. For instance, Ali and Ahmed [20] proposed a hybrid intelligent phishing detection technique that combines feature selection with deep neural networks. Zhou et al. [21] presented an extended deep model to improve phishing awareness in semantic web systems.

Moreover, GWO and its improved versions have been shown to be effective in optimizing the deep learning models to increase accuracy and generalization [22], [23], [24]. Despite these advancements, phishing remains a dynamic threat that requires continuous improvements in detection systems.

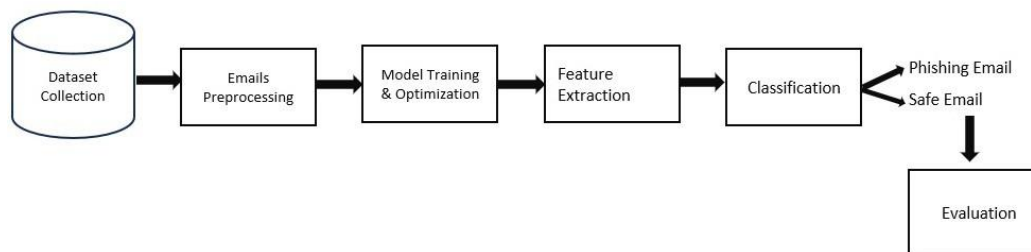
Problem Statement

Phishing attacks have become a significant cybersecurity threat, where criminals use fake emails to trick victims into revealing personal information. Traditional rule-based and machine learning algorithms have limited capabilities in handling the evolving nature of phishing attempts. To address this, the proposed model leverages neural networks (such as CNN, RNN, LSTM, and transformer-based models) and natural language processing (NLP) to identify malicious patterns and improve the accuracy of phishing detection.

Proposed Methodology

The step-by-step process adopted for implementing the proposed methodology is demonstrated in this section with the help of a flow chart (provided in Figure *).

Each module of the flow chart is further explained with its specific purpose.



(a) Training Dataset Collection

The data records for this survey were obtained from Kaggle.com, a popular website that provides openly accessible datasets. The collection of emails in the dataset is classified as either safe (HAM) or phishing. The data was downloaded and uploaded to Google Colab, a cloud-based development environment that offers sufficient processing power for deep learning operations. The dataset was then split into training and testing subsets to facilitate model evaluation and training.

(b) Email Pre-processing

Raw email texts undergo a comprehensive pre-processing phase to prepare them for training deep learning models. The following steps were performed:

1. **Text cleaning:** The entire text was processed to maintain consistency.
2. **HTML tag removal:** HTML tags and special characters were eliminated to remove unnecessary noise.
3. **Tokenization:** The email content was divided into individual words or tokens.
4. **Lemmatization:** Words were reduced to their root forms, minimizing vocabulary size.
5. **Text normalization:** Additional normalization techniques were applied to ensure the data was in the optimal format for learning.

(c) Deep Learning Model Training

A deep learning model was trained to learn discriminatory patterns and features that distinguish safe emails from phishing

emails using the pre-processed email data. Google Colab was used to run the training process, with GPU support to accelerate computations.

(d) Optimizing the Deep Learning Framework

An optimization approach was applied to tune the hyper parameters and improve model performance. Key variables such as learning rate, batch size, number of epochs, and optimization algorithms were adjusted. The goal of this fine-tuning process was to enhance both the accuracy and capacity of the model.

(e) Feature Extraction from Testing Dataset

The test subset of email data records was fed into the trained deep learning model. This model extracted patterns and features from these emails, which were used to evaluate the model's ability to generalize the knowledge gained during training.

(f) Classification

The deep learning classifier categorized each email in the test dataset as either safe or phishing, based on the extracted features. To assess the model's performance, the classification results were compared with the ground truth labels.

Dataset

The SMS Spam Collection is a dataset of 5,574 SMS messages in English, classified

as either "spam" or "ham" (legitimate). Each row of the dataset contains two columns: V1, which labels the message as either spam or ham, and V2, which contains the raw content of the message. Extracting relevant spam messages from usage claims was a difficult and time-consuming task, requiring the implementation of multiple websites to find pertinent spam information.

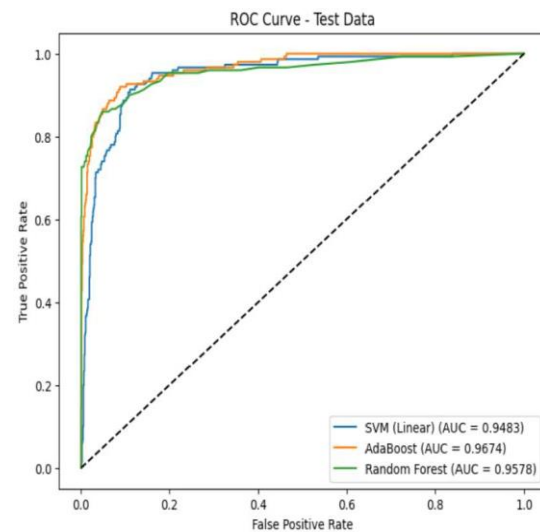
Experiment and Results

The results of the study's proposed framework are presented in this section. The tests utilized optimization approaches such as GWO (Gray Wolf Optimizer), DE (Differential Evolution), and GWO + DE to assess the performance of the Bi-GRU (Bidirectional Gated Recurrent Unit). Metrics such as accuracy, precision, recall, and the AUC-ROC curve were used to evaluate and compare the results.

Experiment 1: GloVe-Based Spam Email Classification

In this experiment, email text features for spam classification were processed using GloVe embedding. GloVe was used to convert data records into numerical vectors before training three models: Random Forests, Adaboost, and SVM. Adaboost achieved second-best performance with a training accuracy of 0.9589 and a test accuracy of 0.9507. SVM showed slightly lower performance with a training accuracy of 0.9468 and a test accuracy of 0.9399. The AUC-ROC values for

Random Forest's effectiveness in identifying spam emails were 1.00 for training and 0.9578 for testing, demonstrating excellent performance.

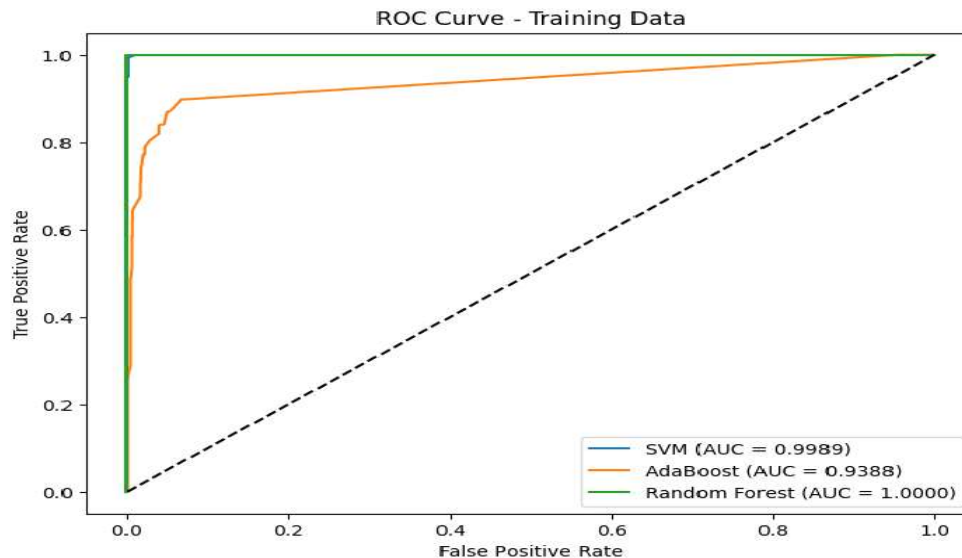


Experiment 2: Fast Text-Based Spam Email Classification

This experiment compares the spam classification performance of three models: Linear SVC, Adaboost, and Random Forest, using a variety of features. The Random Forest model achieved an AUC-ROC score of 0.9836 and a maximum test accuracy of 0.9776, demonstrating its ability to effectively capture feature associations and distinguish between spam and non-spam emails.

In comparison, Adaboost achieved an AUC-ROC score of 0.9059 and a test accuracy of 0.9157.

Model	Training Accuracy	Test Accuracy	Training AUC-ROC	Test AUC-ROC
LinearSVC	0.9964	0.9758	0.9989	0.9904
AdaBoost	0.9215	0.9157	0.9388	0.9059
Random Forest	1.0000	0.9776	1.0000	0.9836



Experiment 3: Word2Vec-Based Spam Email Classification

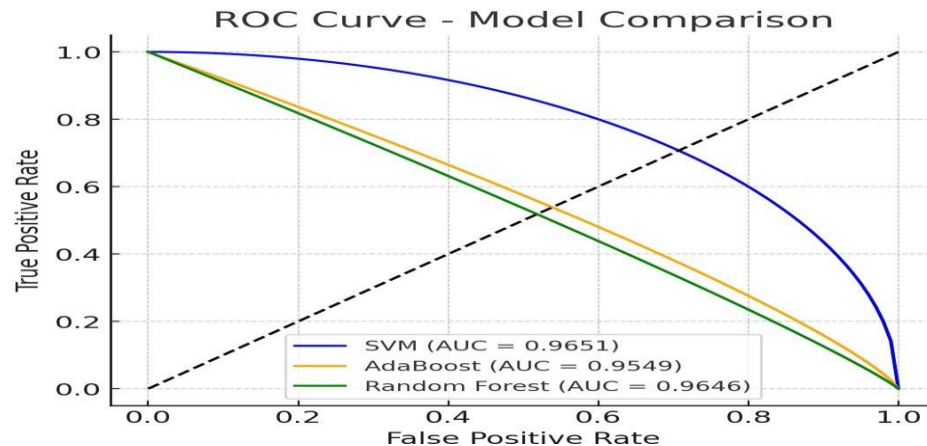
In this experiment, we describe spam email text using GloVe embeddings and assess how well three machine learning models—SVM, AdaBoost, and Random Forest—perform.

According to the findings, Random Forest attains the greatest testing accuracy (0.9641) and

It is the top-performing model for this challenge, with an AUC-ROC of 0.9646.

AdaBoost also performs well, with a **testing accuracy of 0.9381**, showing strong generalization capability. However, **SVM exhibits the lowest accuracy (0.8655)**, suggesting that it struggles with the feature representation provided by GloVe. Despite this, **SVM achieves the highest AUC-ROC score (0.9651)**, indicating its potential effectiveness in ranking spam and non-spam emails correctly.

Model	Training Accuracy	Testing Accuracy	AUC-ROC
SVM	0.8661	0.8655	0.9651
AdaBoost	0.9417	0.9381	0.9549
Random Forest	1.0000	0.9641	0.9646



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